News Framing Digest Bot

AI/ML Engineering · LLM Systems · Backend & Cloud · End-to-End Product Delivery

A production generative-AI system that uses multilingual NLP and a two-model LLM pipeline to show how left, right and international media frame the same news story, delivered daily to Telegram from a serverless, zero-hosting-cost cloud pipeline.

github.com/vishv03/News_Summariser · Live in production

38	9	7	2	$0	$4	151	9
news outlets	languages	world regions	LLMs in pipeline	hosting cost	monthly AI cap	automated tests	design records

Summary

Architected and shipped an **end-to-end AI product**, from data ingestion through machine learning, LLM reasoning and cloud delivery to a live user-facing channel. The system ingests 38 news sources concurrently, applies **transformer-based multilingual embeddings** and **unsupervised clustering** to match coverage of the same event across 9 languages, ranks stories with a custom scoring model, and runs a **cost-optimised LLM pipeline** (Claude Haiku for structured evaluation, Claude Sonnet for generation) to judge political lean, detect loaded language and write neutral comparisons. It runs on **GitHub Actions automation with caching and automated monitoring**, fault-tolerant delivery, and AI spend governed in code. It was built with an **agentic, multi-agent AI development workflow** under my architectural direction.

Core competencies demonstrated

Generative AI & LLM Engineering Multi-model LLM orchestration, structured outputs (JSON schema), prompt & rubric design, LLM evaluation, guardrails	**Machine Learning & NLP** Transformer embeddings, multilingual semantic similarity, unsupervised clustering, zero-shot classification	**MLOps & LLMOps** Model version pinning, embedding caching, CPU-optimised inference, token-level cost observability, budget enforcement
Data Engineering Async ETL from 38 sources, URL canonicalisation & SHA-256 dedupe, schema migrations, data retention policies	**Backend & API Integration** Python services, REST APIs (Anthropic, Telegram, GitHub), async I/O, rate limiting, typed data contracts	**Cloud & DevOps Automation** Serverless GitHub Actions, cron scheduling, concurrency control, build caching, reproducible locked builds
Reliability & Security Idempotent retry-safe delivery, error taxonomy, health monitoring & auto-alerting, least privilege, secret hygiene, supply-chain pinning	**Agentic AI Development** Claude Code with custom subagents, skills and guardrail hooks; AI-assisted review; human-in-the-loop architecture	**Architecture & Leadership** System design, 9 ADRs, trade-off analysis, technical documentation, owning every design decision end to end

Machine learning & NLP

- **Multilingual semantic matching:** encoded articles with a transformer sentence-embedding model (paraphrase-multilingual-MiniLM-L12-v2, sentence-transformers on PyTorch) so coverage of one event in English, French, Arabic, Hindi, Japanese and more is matched **before any paid translation**.
- **Unsupervised story detection:** agglomerative clustering on cosine similarity, with the similarity threshold **tuned and validated on real news data**.
- **Zero-shot topic classification:** designed an embedding-based classifier that assigns each story to its nearest topic description. Ran a comparative experiment on 53 real stories: descriptions misclassified ~3, labels ~6, example headlines ~14, so **error fell about 4x** versus the weakest approach.
- **Custom ranking model:** scores stories by log-damped reach × regional spread × political balance, with per-topic caps for diversity and **automatic "blindspot" detection** for stories covered by only one side or only outside the West.
- **Production ML efficiency:** model pinned to an exact commit for reproducibility, embeddings cached and keyed by model version so only new articles are encoded, and CPU-only PyTorch to fit free CI runners.

Generative AI & LLM engineering

- **Two-model LLM pipeline:** Claude Haiku performs high-volume structured evaluation and Claude Sonnet handles higher-quality generation, a **cost/quality routing** design with per-task reasoning-effort settings.
- **Structured outputs & validation:** JSON-schema responses generated from pydantic models, per-item validation with targeted retries that only fill gaps, and refusal and truncation handling.
- **LLM-as-a-judge evaluation rubric:** scores lean on each outlet's own political axis, requires a justification for every score to reduce anchoring bias, abstains ("insufficient") rather than guessing, and flags **loaded language with neutral alternatives** across languages.
- **LLMOps & cost governance:** every call is costed at token level from an official price table and committed immediately, plus a **pre-flight budget forecast** against a $4/month cap with graceful degradation to a no-AI digest.
- **Responsible AI:** outlet baselines, balanced presentation rules and a dedicated bias-audit agent reviewing every prompt and source change.

Backend, data & cloud engineering

- **Async data ingestion:** concurrent fetching of 38 feeds (asyncio + httpx), HTTP conditional requests (ETag / Last-Modified), response-size limits, URL canonicalisation and SHA-256 deduplication.
- **Data architecture:** split storage, with public state holding only hashes and ids and private article text in a cached store with 48-hour retention. Versioned, atomic schema migrations and automated pruning.
- **Fault-tolerant delivery:** async Telegram Bot API integration with a per-day delivery ledger, so retries never double-post. Errors are handled by type (rate-limit backoff, network retry, no retry on ambiguous timeouts), and messages are paced to platform limits and auto-fit to size limits.
- **Serverless cloud operations:** GitHub Actions with daily and 3-hourly schedules, concurrency groups, model/data/build caching, state persisted even on failure, and **automated alerting**: unhealthy feeds open GitHub Issues that close themselves on recovery.
- **Security:** least-privilege tokens, third-party Actions pinned to commit SHAs (supply-chain hardening), secrets isolated in GitHub Secrets, and logging configured so credentials can't leak.

Agentic, multi-agent AI development

- Led the build as **architect and technical decision-maker**, orchestrating **Claude Code** as an AI engineering partner: I evaluated options and set every threshold, rule, model choice and data format.
- Designed a **multi-agent tooling setup**: custom read-only subagents for a **bias audit** and for **feed health and coverage analysis**, a custom **/adr skill** for decision records, and AI-assisted multi-pass **code review** before every pull request.
- Built **AI guardrails with hooks**: the agent can't commit or push, dependency changes need human approval, and every edited file is automatically formatted and linted. A **human-in-the-loop** workflow that matches how industry teams adopt AI coding agents.

Engineering excellence

- **~5,000 lines of typed Python** with pydantic data contracts between modules, **151 automated tests** using fakes for every external API, static type checking (pyright), linting (ruff), pre-commit hooks, feature branches and reviewed PRs.
- **9 Architecture Decision Records** capturing options, trade-offs and measurable "revisit if" signals, plus a living architecture document with system diagrams.

Technology keywords

AI / ML: Generative AI, Large Language Models (LLMs), Anthropic Claude API, prompt engineering, structured outputs, LLM evaluation, LLM-as-a-judge, NLP, transformers, sentence embeddings, semantic similarity, clustering, zero-shot classification, PyTorch, sentence-transformers, scikit-learn, NumPy, MLOps, LLMOps, responsible AI · **Backend & data:** Python 3.12, asyncio, REST APIs, httpx, pydantic, SQLite, SQL, ETL, data pipelines, schema migrations, Telegram Bot API, RSS/Atom · **Cloud & DevOps:** GitHub Actions, workflow automation, serverless, cron, caching, uv, GitHub CLI, observability, alerting · **Quality & security:** pytest, pyright, ruff, pre-commit, supply-chain security, secrets management · **AI tooling:** Claude Code, AI agents, multi-agent workflows, agentic development

Resume bullets (copy and paste)

- Architected and shipped a production generative-AI system that matches news coverage across 9 languages using transformer embeddings (PyTorch / sentence-transformers) and unsupervised clustering, then runs a two-model Claude LLM pipeline to compare political framing across 38 outlets, delivered daily to Telegram.
- Designed an embedding-based zero-shot topic classifier and validated it experimentally on real data, cutting misclassifications about 4x versus the weakest alternative.
- Engineered LLMOps cost governance with token-level spend tracking, pre-flight budget forecasting and graceful degradation, keeping AI cost under $4/month at $0 hosting.
- Built a fault-tolerant serverless pipeline on GitHub Actions: async ingestion from 38 sources, idempotent retry-safe delivery, automated health monitoring with self-opening GitHub Issues, and SHA-pinned workflows.
- Led development through an agentic, multi-agent AI workflow (Claude Code with custom subagents, skills and guardrail hooks), backed by 151 automated tests and 9 Architecture Decision Records.